137 MHz V-dipole for weather satellites

Two quarter-wave solid copper wires at 120° in a horizontal plane, each laid along a PVC conduit arm in printed clips, fed by an SMA jack in a 3D-printed hub on a PVC mast: the 9A4QV receive antenna for NOAA APT and Meteor LRPT near 137 MHz.

✓ All 18 checks pass

Make it yours

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Centre frequency 137.5 MHz · V angle 120 ° · Copper wire 1.5 mm² solid-core
house wire, ø1.4 mm copper · Insulation Stripped bare (the length is
exact) · Mast pipe PVC-U pressure pipe, 40 mm outside, PN16 · Design wind
gust 35 m/s gusts (exposed roof or garden) · Mast pipe length 1000 mm ·
Coax length 3000 mm · Choke Five turns of the coax round the mast
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OVERALL	MASS	PIECES	WIND ON MAST AND ANTENNA HEAD
942 × 341 × 1034 mm length × depth × height	**1.0 kg** as built, with every part listed	**5 cut (3 different)**	**factor 2.5** safety factor on yield

COAX SNAGGED AT THE HUB	3D PRINTS	BUILD TIME	COST
factor 4.0 safety factor on yield	**10 parts** about 142 g of filament, 12.5 h printing	**7.5 h** 9 steps	**~$51** everything on the materials list

FREQUENCY	ELEMENT	V ANGLE	TIP TO TIP
137.5 MHz centre; NOAA APT and Meteor LRPT	**531 mm** each, 1.5 mm² copper, from its inner end	**120°** opening north or south	**941 mm** as cut

ARMS	CLIPS	HUB HEIGHT
16 mm PVC conduit rigid electrical conduit	**6** 3 per arm	**1.03 m** above the bottom of the mast

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Checks

All 18 checks pass at this size.

	Check	Result
✓	**Every print fits the printer** As it sits on the bed, turned the way the files come.	`Feed hub: 99 × 104 × 81 on a 220 × 220 × 250 bed`
✓	**Nothing floating** Every member, plate and part touches what holds it.	`all connected`
✓	**No clashes** No member passes through another away from a joint, and no plate through a member it is not notched around.	`none`
✓	**Every piece fits the stock** Cut from standard lengths with a trim and a kerf per cut.	`longest 1000 mm`
✓	**Every piece can be clamped and cut** No piece shorter than 32 mm on its short side.	`shortest 497 mm`
✓	**Wind on mast and antenna head: strength** Yield ÷ peak stress with 51 N of drag in 35 m/s gusts (27 N on the mast, 24 N on the hub, arms, wires, clips and caps, at the top), plus self weight.	`safety factor 2.54 (need 2), worst C (mast)`
✓	**Coax snagged at the hub: strength** Yield ÷ peak stress with 25 N pulled sideways at the hub, plus self weight.	`safety factor 3.99 (need 2), worst C (mast)`
✓	**Every part has a step** Each piece is cut, fitted or bought in a numbered step.	`9 steps`
✓	**Element length from frequency** Measured from the wire's inner end in the hub. End effect for a ⌀1.4 mm conductor 1.6 %; the PVC arm and clips touching it about 1 %; bare copper. Cut 1 % long and trimmed at the tip in step 8.	`λ/4 at 137.5 MHz = 545 mm; × 0.984 × 0.99 = 531 mm per wire; cut 536 mm`
✓	**V angle** At 120° the horizontal V has its most gain overhead and a usable lobe down to low passes, and its impedance comes close to 50 Ω. Narrower drops the impedance; 180° is a plain dipole with nulls off its ends.	`120° between the elements`
✓	**Feed gap** Small against the wavelength so it does not detune the feed, and wide enough that the two clamps cannot touch (at least 4 mm).	`12 mm between the wire ends (0.55 % of a wavelength); 13 mm between the two sides' washers and wires`
✓	**Feed match (rough estimate)** Estimate R ≈ 73 Ω × $\sin^2$(angle ÷ 2) at resonance; the NanoVNA in step 8 gives the real number. Under 2 is plenty for receiving.	`about 55 Ω, SWR 1.09 on 50 Ω coax`
✓	**Choke on the coax** At least 500 Ω of common-mode impedance keeps current off the outside of the shield, so the coax does not become part of the antenna and pick up house noise. Coil from Wheeler's formula; ferrites from the datasheet curve.	`5 turns, about 1.10 µH, 951 Ω at 137.5 MHz`
✓	**Coax loss** Under 3 dB keeps the weak satellite signal above the receiver noise; use a better cable (or a preamp at the hub) for a longer run.	`0.8 dB over 3.00 m of RG58 with its plugs`

✓	**Wire held along the arm** Held at the screw, through the hub channel, by a clip at least every 120 mm and by the tip cap, so the soft wire stays straight on the arm; the tip past the cap carries its own wind and weight against annealed copper (70 MPa).	3 clips per arm every 114 mm, longest run between holds 109 mm; tip 17 mm past the cap, safety factor 96.41, sags 0.0 mm
✓	**Arms in the wind** Each 16 mm conduit arm as a cantilever out of the hub in 35 m/s gusts, with the wire, clips and cap on it and their weight, against PVC's long-term 25 MPa. The design uses the thinner conduit where it holds and the thicker one where it does not.	safety factor 2.44 (need 2); each arm 476 mm out of its socket
✓	**Printed hub fits** Push fits 0.4 mm over the size, at least 3 mm of wall round a socket that is clamped, the wire standing proud by a ring terminal's thickness so the washer clamps both; fits a 200 mm bed.	mast socket ø40.4 mm on ø40 mm pipe; arm sockets ø16.4 mm on ø16 mm conduit, 26 mm deep, 3.5 mm wall; wire channel ø1.8 mm, groove ø1.7 mm with the wire 0.8 mm proud; 99 × 104 on the bed
✓	**Clips and caps fit** The mouth is 75 % of the tube, so the C ring opens 4 mm to pass it: under 2 % strain in PETG or ASA, which springs back. 3 mm ring walls; at least 1.2 mm (three perimeters) over the wire hole.	clip bore ø16.2 mm snug on ø16 mm, mouth 12 mm, springs open 1.4 % to snap on; cap socket ø16.3 mm; wire hole ø1.8 mm for ø1.4 mm copper, 2 mm wall over it

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2 Materials

Stock

Section	Pieces	Cut length	Buy	Mass	Cost
16 × 1.4 rigid PVC electrical conduit	2	1.0 m	1 × 3.00 m	0.1 kg	$3.00
1.5 mm² solid copper house wire (ø1.4 conductor)	2	1.1 m	2 × 1.00 m	0.0 kg	$1.20
40 × 3 PVC-U pressure pipe, PN16	1	1.0 m	1 × 3.00 m	0.5 kg	$14

3D prints

Part	Settings	Qty	Filament	Print	Step	Cost	Files
Feed hub	PETG · 0.2 mm layers · 4 walls · 40% infill · no supports	1	121 g	10.1 h	1	$2.66	
Pocket lid	PETG · 0.2 mm layers · 4 walls · 40% infill · no supports	1	6 g	0.5 h	1	$0.14	
Wire clip	PETG · 0.2 mm layers · 4 walls · 60% infill · no supports	6	7 g	1.2 h	5	$0.15	
Tip cap	PETG · 0.2 mm layers · 4 walls · 60% infill · no supports	2	8 g	0.7 h	5	$0.18	

Filament and time are estimates from the part's volume; your slicer has the real numbers. The .stl is the part at this size, in millimetres; the .scad is its OpenSCAD source.

Parts to buy

Part	Spec	Qty	Step	Cost
Heat-set insert, M3	M3, short (about 4 long), for a 4.0 hole	6	1	$0.60
SMA female bulkhead jack, solder cup	panel mount, 50 Ω, with nut and lock washer, for panels up to 4 thick	1	2	$2.00
Solder tag to fit the SMA jack thread	tinned brass ground lug	1	2	$0.20
Ring terminal, M3	tinned copper, for 0.5 to 1.5 square wire, solder or crimp	2	2	$0.20
Hook-up wire	stranded, silicone insulated, 0.5 square	1	2	$0.50
Grub screw, M3 × 8, cup point, stainless	into the inserts under the sockets, onto the arms	2	3	$0.20
Screw, M3 × 8, pan head, stainless	clamps each wire end into its insert in the pocket floor	2	4	$0.20
Penny washer, M3 × 12, stainless	3.2 hole, 12 outside, about 1 thick	2	4	$0.20
Bolt, M4 × 60, stainless, with nyloc nut and two washers	through the hub socket and the mast	1	6	$0.40
Coax, RG58, SMA male to SMA male	3.0 metre assembly (or the plug your receiver needs at the far end)	1	7	$9.60
Cable ties, UV-stable black	200 long	9	7	$0.45
Screw, M3 × 8, pan head, stainless	lid into the heat-set inserts	2	9	$0.20
Mast clamp (U-bolt with saddle)	M6 U-bolt for a 40 pipe, with a saddle for your post or wall bracket	2	9	$6.00

Consumables

- Solder, rosin core · a little
- Self-amalgamating tape · 1 roll
- PVC electrical tape, black · 1 roll
- Neutral-cure silicone sealant · 1 small tube

Tools

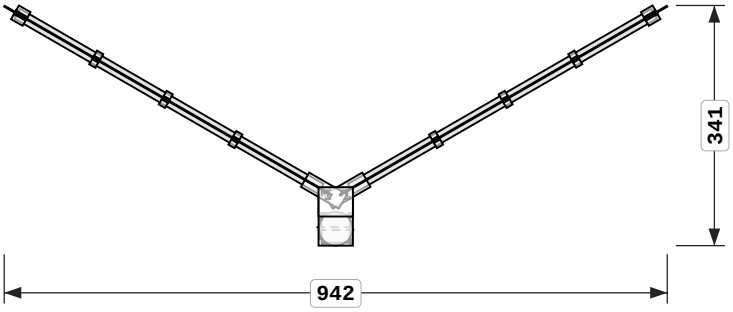
- 3D printer and the filament named with each print
- Soldering iron (with a heat-set insert tip if you have one)
- Fine-tooth saw, file and deburring tool
- Drill and a 4.3 mm bit
- Vise, pliers, side cutters, wire strippers and a sharp knife
- Calipers and tape measure
- Multimeter
- NanoVNA or antenna analyser
- Spanner for the SMA nut, screwdrivers, hex key for the grub screws
- Spirit level and compass

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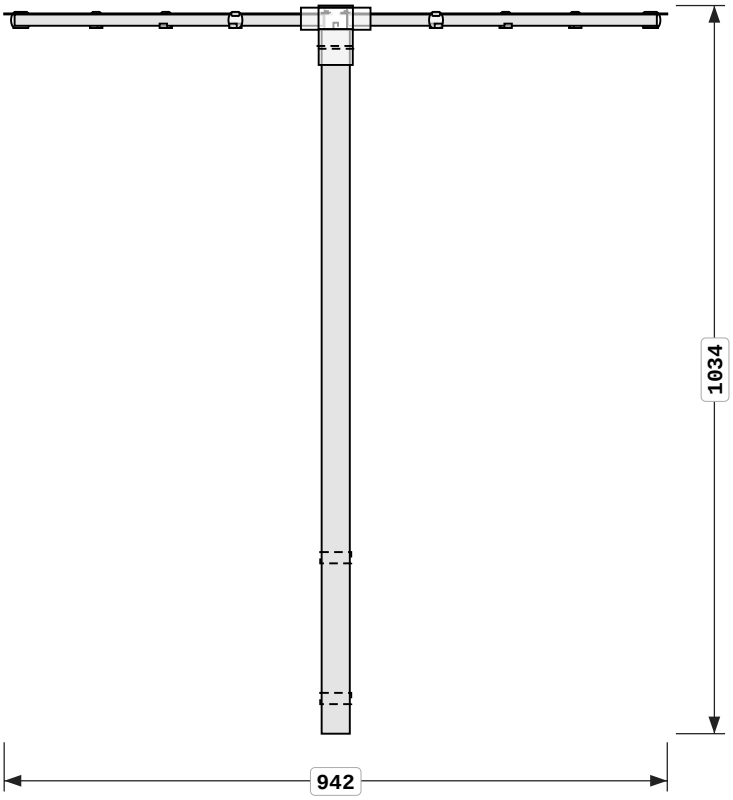
3 Drawings

Overall dimensions. Every number on this page is from the same model as the 3D view.

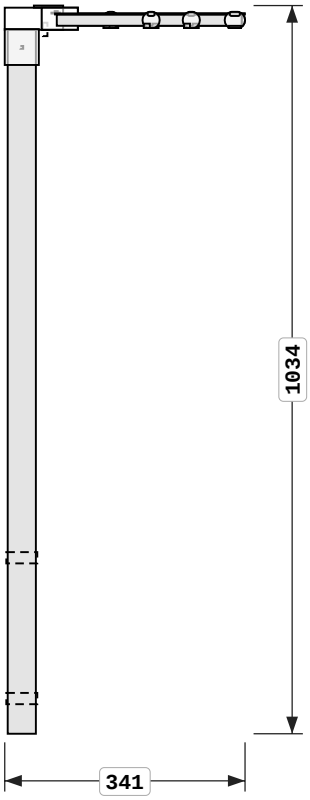
Top



Front



Side (from the right)



4 Legend

Units	millimetres, kilograms, degrees
Datum	x east from the apex of the V (the feed point), y north (the V opens toward north), z up from the bottom end of the mast
Piece marks	A, B, C… one letter per different piece; x4 after it means cut four the same. Mirror images share a mark.
Cut marks	From one square end, mark each edge at the two numbers, join the marks around the section, cut on the lines. Angles are for reference.
Edges	named by where they face on the finished build (top-south, bottom-east…)
Joints	nothing is welded: parts clamp, as each step says
Long point	the overall length of the piece, tip to tip
Dimensions	jig drawings and joint tables give centre lines (℄), where the members' centres cross; the general arrangement gives outside sizes
Tolerance	cut ±1 mm unless a step says otherwise

5 Order

Start the hub print, then wire the jack and cut the arms while it runs; clamp the wires, clip them to the arms, fit the mast, tune by trimming the tips, then seal and mount.

Fabricate

1. Print the hub and the lid 1 h
2. Wire the SMA jack 0.5 h
3. Arms 0.5 h
4. Element wires: cut, strip and clamp 1 h
5. Clips and tip caps 0.5 h
6. Mast and hub 1 h

Assemble

7. Coax and coil choke 0.5 h

Tune and install

8. Tune with a NanoVNA 1 h
9. Seal, mount and point it 1.5 h

6 Steps

Ticks save on this device. "Show in 3D" shows the build after that step, the new parts lit.

01 Print the hub and the lid

Fabricate · 1 h

- [ ] Print the feed hub and the pocket lid from the files below (settings and orientation with each), then queue the 6 wire clips and 2 tip caps from step 5. PETG or ASA, not PLA: it lives outdoors.
- [ ] Start the hub first: it takes a few hours, and steps 2 and 3 can start while it runs.
- [ ] Clean the strings out of the arm sockets, the wire channels, the SMA hole and the mast socket. Push a scrap of the wire through each channel: it must slide.
- [ ] Melt an M3 heat-set insert with a soldering iron into each of the six 4 mm holes, flush with the face: two in the pocket floor (wire clamps), two from underneath into the arm sockets (grub screws), two on the top face behind the pocket (lid).

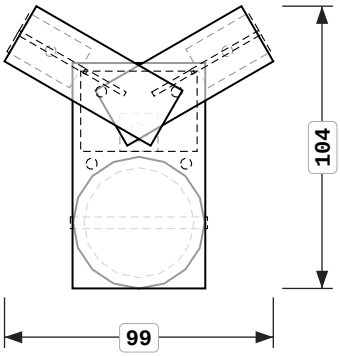
Check A conduit offcut pushes into each arm socket to the bottom by hand; the mast pipe pushes into its socket to the end; all six inserts square and flush within ±0.5 mm.

> A socket that came out tight is opened with a file or a drill turned by hand, never forced: forcing the tube in splits the hub along its layers.

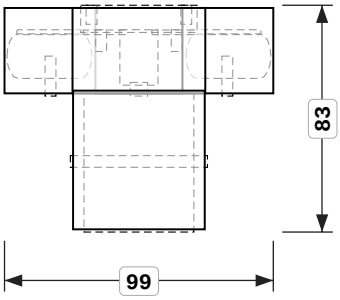
Print: Feed hub

PETG · 0.2 mm layers · 4 walls · 40% infill · no supports · about 121 g, 10.1 h each · print it top face down on the bed, mast socket pointing up (the files come turned this way); no supports: the pocket floor and the tops of the arm bores bridge

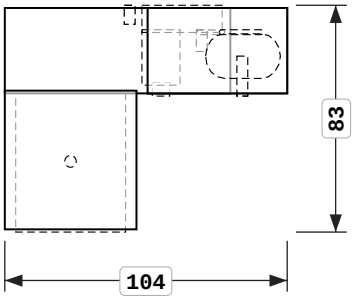
Feed hub: Top



Feed hub: Front



Feed hub: Side (from the right)



Cut	Size	Axis	From (x, y, z)	Depth
Hole	ø40.4	z	49.4, 24.2, -1	51
Hole	ø4.3	x	24.2, 24.2, 25	50.4
Pocket	42.7 × 29.5 × 10		49.4, 65.2, 77.4	
Pocket	14 × 14 × 20.4		49.4, 57.4, 63.2	
Hole	ø6.4	z	49.4, 57.4, 49	5
Hole	ø16.4	angled	70.8, 80.9, 63.7	26.6
Hole	ø16.4	angled	28, 80.9, 63.7	26.6
Hole	ø1.8	angled	68.7, 79.7, 72.5	29.1
Hole	ø1.8	angled	30.2, 79.7, 72.5	29.1
Hole	ø1.7	angled	54.6, 71.5, 72.5	17.3
Hole	ø1.7	angled	44.3, 71.5, 72.5	17.3
Hole	ø4	z	63.3, 72.4, 65.4	8
Hole	ø4	z	35.6, 72.4, 65.4	8
Hole	ø4	z	81.9, 87.3, 49	14.7
Hole	ø4	z	16.9, 87.3, 49	14.7
Hole	ø4	z	32.1, 45.9, 75.4	7
Hole	ø4	z	66.8, 45.9, 75.4	7

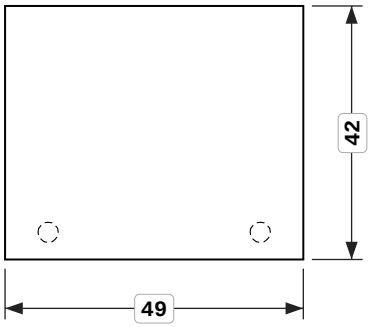
Dashed on the drawing. Positions from the part's lower-left-bottom corner as drawn; the files come turned for printing (+z face on the bed). On the bed it is 98.8 × 103.8 × 81.4 mm.

> Melt an M3 heat-set insert into each of the six small holes: two in the pocket floor, two under the arm sockets, two on the top face behind the pocket.

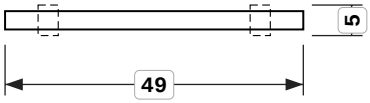
Print: Pocket lid

PETG · 0.2 mm layers · 4 walls · 40% infill · no supports · about 6 g, 0.5 h each · print it flat on the bed

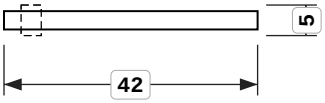
Pocket lid: Top



Pocket lid: Front



Pocket lid: Side (from the right)



Cut	Size	Axis	From (x, y, z)	Depth
Hole	ø3.3	z	7, 4.5, -1	5
Hole	ø3.3	z	41.7, 4.5, -1	5

Dashed on the drawing. Positions from the part's lower-left-bottom corner as drawn. On the bed it is 48.7 × 41.5 × 3 mm.

> Screwed down over a bead of neutral-cure silicone after tuning.

Parts

- Heat-set insert, M3 ×6 M3, short (about 4 long), for a 4.0 hole

02 Wire the SMA jack

Fabricate · 0.5 h

- [ ] Cut two tails of hook-up wire 50 mm long, strip 4 mm at each end, and solder an M3 ring terminal on one end of each.
- [ ] Slip the solder tag over the thread of the SMA jack. Solder one tail to the centre solder cup (east wire) and the other to the solder tag (shield, west wire).
- [ ] Push the jack, thread first, down through the hole in the floor of the small well behind the pocket, so it points down under the hub; fit the lock washer and nut from underneath and tighten with a spanner while holding the jack body: it must not turn.
- [ ] Lead the centre tail to the east clamp insert (right, looking north) and the tag tail to the west one, ring terminals over the inserts.

Check Multimeter: centre pin to jack body open circuit (no short); centre pin to the east ring terminal and jack body to the west ring terminal both under 1 Ω.

> Heat the solder cup briefly, and solder before the jack goes in the hub: the jack's insulator and the PETG both soften if the iron stays on.

Parts

- SMA female bulkhead jack, solder cup x1 panel mount, 50 Ω, with nut and lock washer, for panels up to 4 thick
- Solder tag to fit the SMA jack thread x1 tinned brass ground lug
- Ring terminal, M3 ×2 tinned copper, for 0.5 to 1.5 square wire, solder or crimp
- Hook-up wire x1 stranded, silicone insulated, 0.5 square

03 Arms

Fabricate · 0.5 h

- [ ] Cut the two arms (A) from the 16 mm conduit with a fine saw, square, to the length on the cut sheet; deburr inside and out.
- [ ] Push each arm into its socket on the hub until it bottoms.
- [ ] Turn a grub screw into each insert under the sockets until it bears on the arm, then a quarter turn more: snug, not cracking the tube.

Check Both arms bottomed in their sockets; arm tip to arm tip 904 ±5 mm; neither arm turns or pulls out by hand; with the hub on a flat table both arm tips the same height within 3 mm.

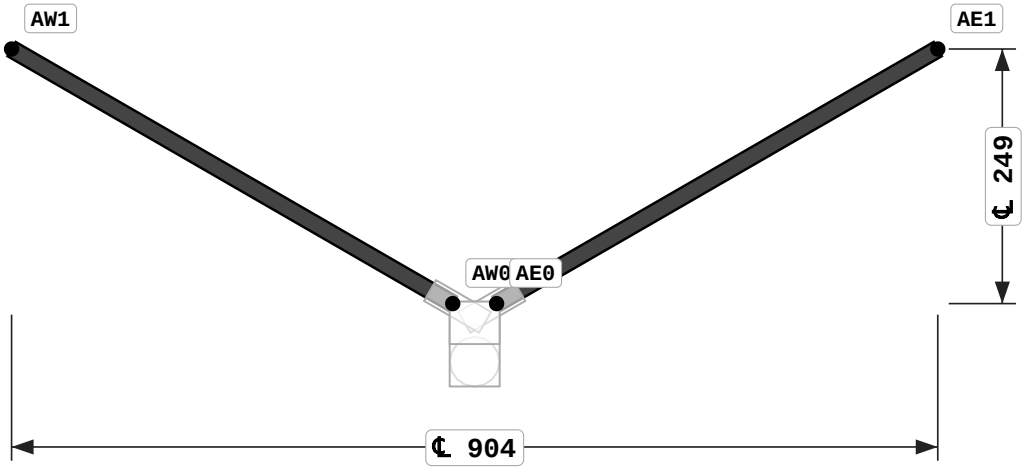
> The arms only carry the wire and the wind on it; the clips and the tip caps follow the arm, so an arm that sits askew puts the wire askew.

Parts

- Grub screw, M3 × 8, cup point, stainless x2 into the inserts under the sockets, onto the arms

Jig

Step 3: Top



Joint positions

Point	x	y	z
AW0	-21	12	1014
AE0	21	12	1014
AW1	-452	261	1014
AE1	452	261	1014

Pieces: A x2

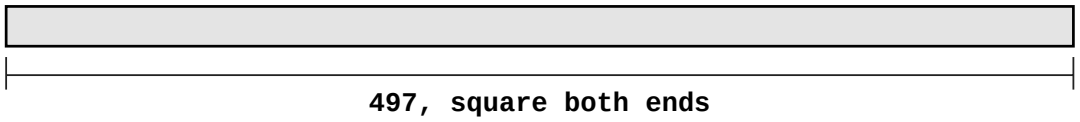
A x2

497 mm

Arm · Conduit 16 · step 3

long point

`arm-E, arm-W`



Left end: free end · Right end: free end

04 Element wires: cut, strip and clamp

Fabricate · 1 h

- [ ] Pull a length of the 1.5 mm² wire straight: clamp one end in a vise and pull the other with pliers until it stretches slightly and lies dead straight. Then cut the two element wires (B) to the length on the cut sheet, square, with side cutters.
- [ ] Strip all the insulation off: slit it along its length with a sharp knife held flat and peel it off, without nicking the copper. Wipe the copper clean.
- [ ] Feed the inner end of each wire into the channel at the outer end of the hub arm, along the top of the arm, until it comes out in the pocket and lies in the groove in the pocket floor.
- [ ] Slide the wires until their inner ends are 12 mm apart (calipers across the pocket).
- [ ] Over each insert: the ring terminal (centre tail on the east, tag tail on the west), then the penny washer over the terminal and the wire, then the M3 × 8 screw. Tighten until the washer bites the wire flat in its groove.

Check Inner ends 12 ±1 mm apart; each wire stays put when you tug it firmly; multimeter: SMA centre pin to the east wire and jack body to the west wire both under 1 Ω, and centre pin to jack body still open circuit.

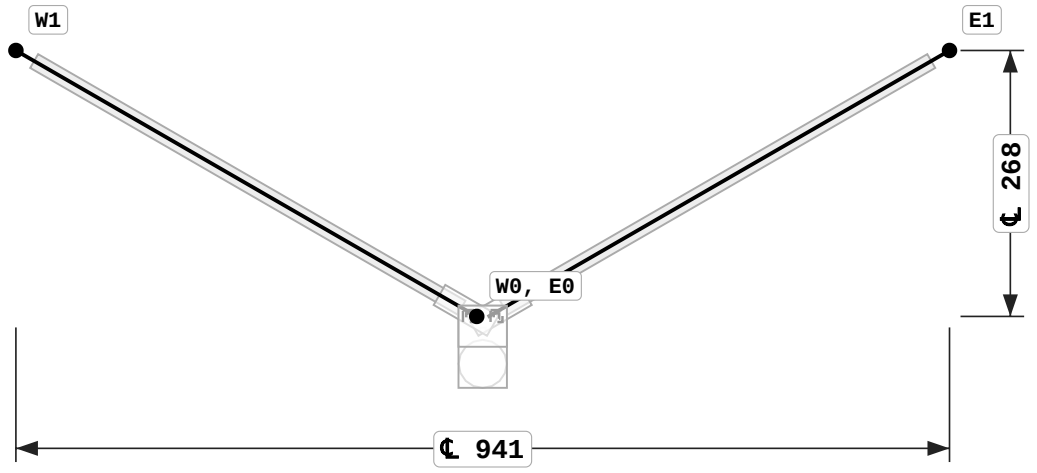
> The element length is measured from the wire's inner end, so set the 12 mm gap before you tighten. The cut sheet length is 5 mm over the design length on purpose: tuning in step 8 trims it.

Parts

- Screw, M3 × 8, pan head, stainless x2 clamps each wire end into its insert in the pocket floor
- Penny washer, M3 × 12, stainless x2 3.2 hole, 12 outside, about 1 thick

Jig

Step 4: Top



Joint positions

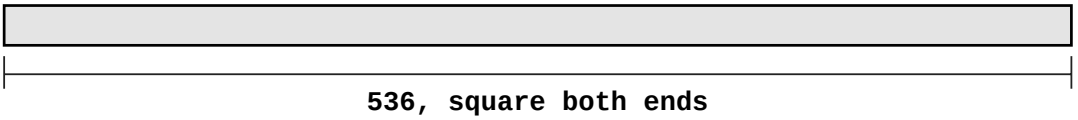
Point	x	y	z
W0	-6	3	1022
E0	6	3	1022
W1	-470	272	1022
E1	470	272	1022

Pieces: B x2

B x2 — **536 mm**

Element wire · Cu wire 1.5 · step 4 — long point

`wire-E, wire-W`



Left end: free end · Right end: free end

05 Clips and tip caps

Fabricate · 0.5 h

- [ ] Print the 6 wire clips and 2 tip caps (files below) if they are not done yet. Clean the wire hole of each with a scrap of the wire, and snap one clip onto a conduit offcut: it must click on and grip.
- [ ] From the tip end, thread 3 clips onto each wire, mouths down, then a tip cap, socket toward the hub.
- [ ] Push each tip cap over the end of its arm until the tube bottoms in the cap, with the wire coming straight out of the end wall; a dab of neutral-cure silicone in the socket keeps it on.
- [ ] Lay the wire straight along the top of the arm. Mark the clip centres on each arm, measured from the end face of the hub's arm socket (table below), and snap each clip down onto the arm on its mark, the wire through its hole.
- [ ] Pull the wire gently from the tip so it lies straight on the arm from the hub to the cap, and leave the tip standing 17 mm out past the cap.

Clip centres along each arm

Clip	From the hub arm end
Clip 1	114
Clip 2	228
Clip 3	342

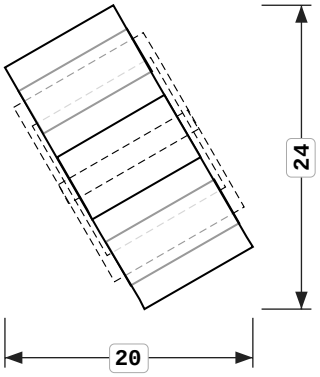
Check Each wire lies on top of its arm all the way, no kinks; clips on their marks within 5 mm; each tip stands 17 ±3 mm past its cap; wire tip to wire tip 941 ±5 mm.

> The clips hold the wire on the arm and slide along it; they do not have to clamp it, since the screw in the hub holds the wire's end.

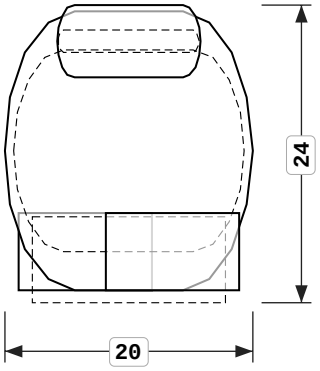
Print: Wire clip x6

PETG · 0.2 mm layers · 4 walls · 60% infill · no supports · about 1 g, 0.2 h each · print it as modelled: lying on its side, the open mouth down, standing on its two flat feet; no supports (the top of the bore bridges)

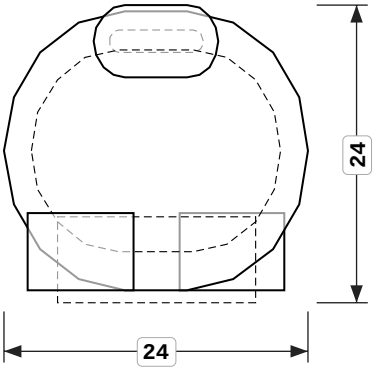
Wire clip: Top



Wire clip: Front



Wire clip: Side (from the right)



Cut	Size	Axis	From (x, y, z)	Depth
Hole	⌀16.2	angled	4.7, 9.2, 11.2	12
Hole	⌀1.8	angled	4.7, 9.2, 20	12
Cut	prism			

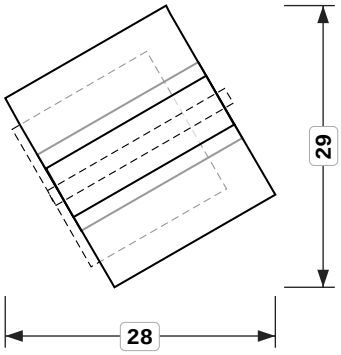
Dashed on the drawing. Positions from the part's lower-left-bottom corner as drawn. On the bed it is 19.9 × 24.4 × 22.9 mm.

> Thread 3 onto each wire before the tip cap; snap onto the arm from above.

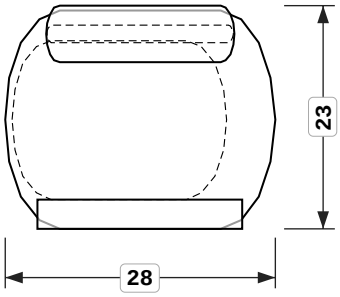
Print: Tip cap x2

PETG · 0.2 mm layers · 4 walls · 60% infill · no supports · about 4 g, 0.3 h each · print it as modelled: lying on its flat foot, the socket and end wall standing up; no supports (the top of the socket bridges)

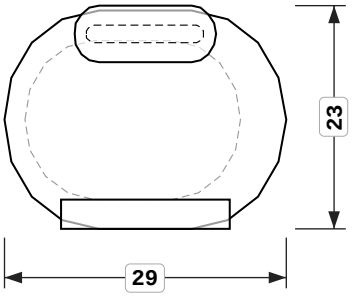
Tip cap: Top



Tip cap: Front



Tip cap: Side (from the right)



Cut	Size	Axis	From (x, y, z)	Depth
Hole	ø16.3	angled	4.7, 9.2, 11.2	16
Hole	ø1.8	angled	4.7, 9.2, 20	21

Dashed on the drawing. Positions from the part's lower-left-bottom corner as drawn. On the bed it is 27.7 × 28.9 × 22.9 mm.

> Pushed over the end of the arm; the wire comes out through the end wall.

06 Mast and hub

Fabricate · 1 h

- [ ] Cut the mast (C) to length from the 40 mm PVC pipe with a fine saw, square; deburr inside and out.
- [ ] Mark a line 25 mm down from the top end and drill a 4.3 mm hole straight through the pipe on it, across a diameter.
- [ ] Push the hub onto the top of the mast until the pipe bottoms in the socket, turn it until the cross holes line up, and fit the M4 bolt with a washer each side and the nyloc nut.
- [ ] Mark the clamp positions on the mast: 50 and 250 mm up from the bottom end.

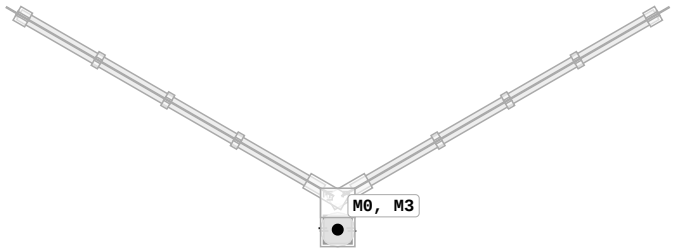
> **Check** Mast within ±1 mm of its length; the hub does not turn or lift on the mast by hand; both arms level when the mast is held plumb (a spirit level on each arm).

Parts

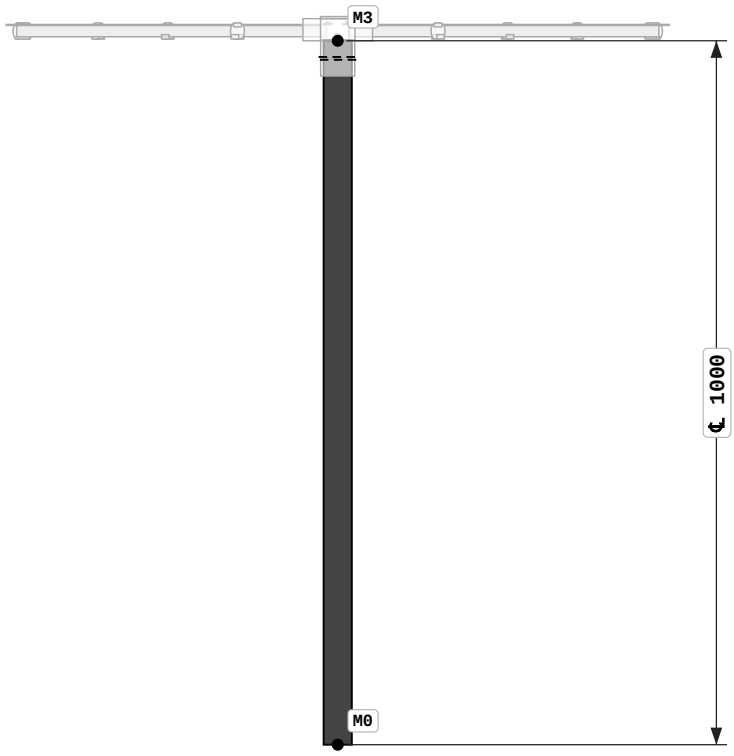
- Bolt, M4 × 60, stainless, with nyloc nut and two washers x1 through the hub socket and the mast

Jig

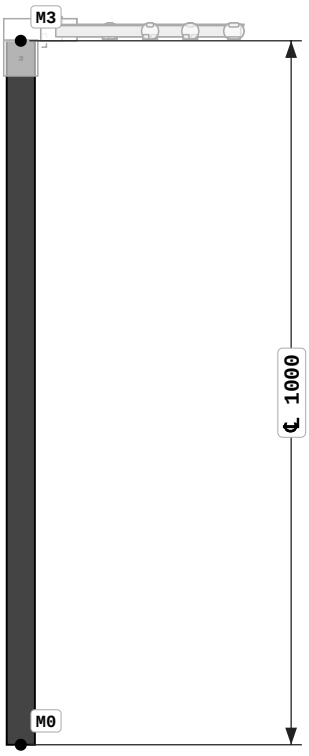
Step 6: Top



Step 6: Front



Step 6: Side (from the right)



Joint positions

Point	x	y	z
M0	0	-44	0
M3	0	-44	1000

Pieces: C x1

C x1

1000 mm

Mast · PVC 40×3 · step 6

long point

mast



1000, square both ends

Left end: free end · Right end: free end

07 Coax and coil choke

Assemble · 0.5 h

- [ ] Screw the SMA plug of the coax onto the jack under the hub, finger tight, then an eighth of a turn with a spanner.
- [ ] Wrap the SMA joint in self-amalgamating tape, stretched, half lapped, from the hub underside down onto the cable; then a layer of PVC tape over it against UV.
- [ ] Directly under the hub socket, wind 5 tight turns of the coax round the mast, side by side (about 25 mm of mast), and hold the coil with tape at each end.
- [ ] Run the coax down the mast and cable-tie it every 200 mm; leave a drip loop where it leaves the mast.

Check 5 turns, tight against each other and against the mast, starting within 50 mm of the hub; nothing slides when you tug the cable.

Parts

- Coax, RG58, SMA male to SMA male x1 3.0 metre assembly (or the plug your receiver needs at the far end)
- Cable ties, UV-stable black x9 200 long

08 Tune with a NanoVNA

Tune and install · 1 h

- [ ] Clamp the mast upright in the open, elements at least 1.5 times their own length from walls, metal and people (a temporary post, a fence, a tripod).
- [ ] Calibrate the NanoVNA (open, short, load) at the end of its own short lead, then connect the far end of the antenna coax.
- [ ] Sweep 120 to 160 MHz and read the frequency of the lowest SWR. The aim is 137.5 MHz ±1 MHz; as cut, the wires are long, so expect the dip a little low.
- [ ] If the dip is low, the wires are long: snip the same amount off the tip of each wire with side cutters, about 3.9 mm for each MHz, and sweep again. Snip less than you think, a few millimetres at a time.
- [ ] If the dip is high, the wires are short: leave them if SWR at 137.1 and 137.9 MHz is under 2; otherwise cut a new pair 3.9 mm longer for each MHz it is high (the clips and caps slide off and go back on).
- [ ] No NanoVNA? An SWR meter needs a transmitter, and you may not transmit at 137 MHz. A licensed operator can read SWR at 144 and 146 MHz instead: it should be low at 144 and rise toward 146.

Check SWR minimum within ±1 MHz of 137.5 MHz and SWR under 2 from 137.1 to 137.9 MHz; both wires stand out past their caps by the same amount within ±1 mm.

> Trim both wires equally and a little at a time: a millimetre is easy to take off and impossible to put back. Bend nothing at the tips: a hooked tip changes the length.

09 Seal, mount and point it

Tune and install · 1.5 h

- [ ] Fill the pocket with neutral-cure silicone (not acetic: it corrodes the joints) over the screws, the wire ends and the jack, run a bead where each wire enters its channel, lay the lid on, and screw it down with the two M3 × 8 screws.
- [ ] Fix the mast to its post, rail or wall bracket with the two U-bolt clamps on the marks, 50 and 250 mm up from the bottom end. Both clamps tight: the wind check counts on them.
- [ ] Turn the mast so the V opens toward north or south (true north: correct a compass for the local declination). East-west passes are what the open side faces away from.
- [ ] Set the mast plumb so both arms are level, and put it where it sees as much of the sky as you can, above the roof line if possible and away from other antennas.
- [ ] Run the coax to the receiver with the drip loop below the entry point.

Check Opening within 10° of north or south; each arm level within 3° (a spirit level bubble inside the lines); the mast does not move when you shake it at the hub.

> Mount it lower rather than in a place where the mast cannot be clamped at both marks: the wind check assumes two tight clamps on a post that does not move.

Parts

- Screw, M3 × 8, pan head, stainless x2 lid into the heat-set inserts
- Mast clamp (U-bolt with saddle) x2 M6 U-bolt for a 40 pipe, with a saddle for your post or wall bracket

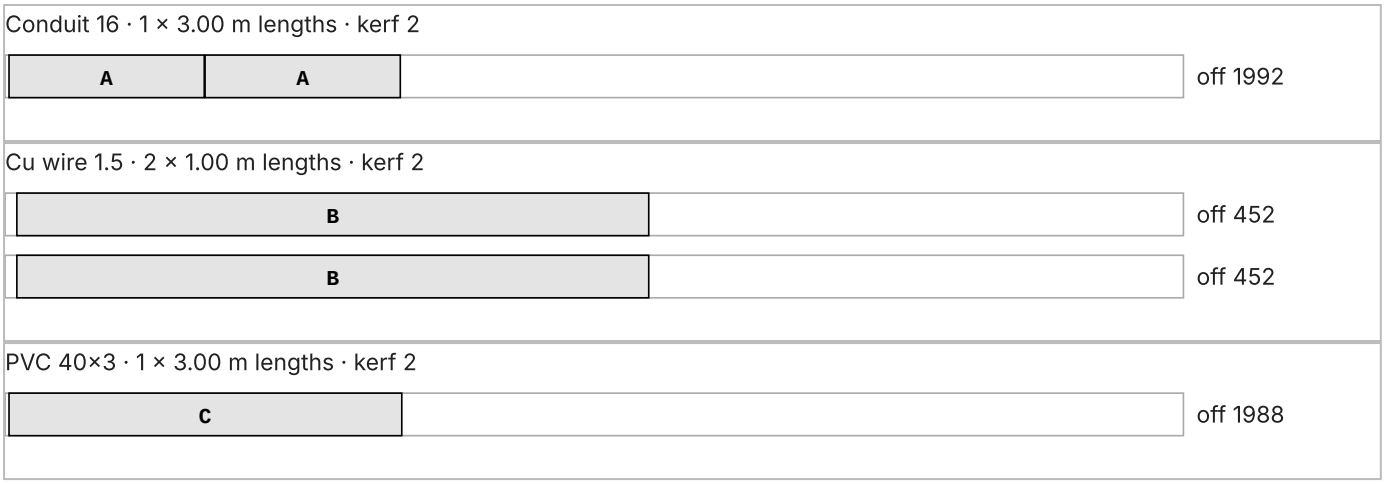
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7 Cut list

Mark	Qty	Section	Long point	Left end	Right end	Step
A Arm	2	Conduit 16	497	square · free end	square · free end	3
B Element wire	2	Cu wire 1.5	536	square · free end	square · free end	4
C Mast	1	PVC 40×3	1000	square · free end	square · free end	6

Cutting plan

Each bar is one stock length; cut the pieces left to right.



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137 MHz V-dipole for weather satellites · v2. Generated by the blueprint skill. Check every dimension against your own stock before you cut, wear eye and hand protection, and load-test the build gently before trusting it.